

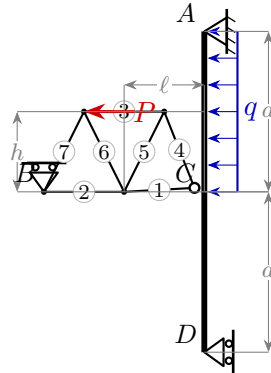
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Exam **Group A**

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Model solution



1. Static determinacy

Counting criterion $n = 3k - (a + z)$ (bodies k , reactions a , interaction forces z):

$$n = 3k - (a + z) = 3 \cdot 2 - (4 + 2) = 0 \Rightarrow \text{statically determinate}$$

2. Support reactions and hinge forces

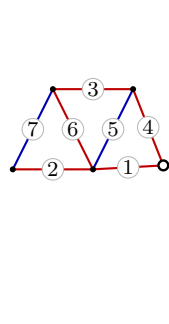
A : $R_x = 14.25$, $R_y = -7.5$ D : $R_x = 9.75$, $R_y = 0$ B : $R_x = 0$, $R_y = 7.5$

C : $H_x = -15$, $H_y = 7.5$

3. Truss member forces

Member	Force S [kN]	Type
1	11.25	tension
2	3.75	tension
3	7.5	tension
4	8.39	tension
5	-8.39	compression
6	8.39	tension
7	-8.39	compression

red: tension (+) blue: compression (-)



4. Section-force diagrams of the beams